

## SDG Un 3

# Analyze how Naïve Bayes can classify recyclable and non-recyclable waste using sensor data

## Introduction

Proper waste management is one of the biggest environmental challenges faced by modern cities. Every day, large amounts of plastic, paper, glass, metal, food waste, and electronic waste are generated. If recyclable and non-recyclable waste are mixed together, recycling becomes difficult, resulting in increased landfill waste and environmental pollution. Manual waste segregation is time-consuming, expensive, and prone to human error. Therefore, smart waste management systems that use machine learning techniques have become increasingly important.

One of the simplest and most effective machine learning algorithms for waste classification is **Naïve Bayes**. This algorithm uses probability to predict the category of an object based on the information collected from different sensors. It can quickly determine whether a waste item is recyclable or non-recyclable by analyzing sensor readings such as weight, moisture level, metal detection, and color.

This approach supports **Sustainable Development Goal (SDG) 11 – Sustainable Cities and Communities** by promoting cleaner urban environments and **SDG 12 – Responsible Consumption and Production** by encouraging efficient recycling and reducing waste sent to landfills.

## Naïve Bayes Algorithm

Naïve Bayes is a **supervised machine learning classification algorithm** based on **Bayes' Theorem**. It predicts the probability that a given object belongs to a particular class.

The algorithm is called "Naïve" because it assumes that all input features are independent of one another. Although this assumption may not always be true in real-world situations, the algorithm performs remarkably well for many classification problems.

The mathematical formula of Bayes' Theorem is:

$$P(C|X) = \frac{P(X|C) \times P(C)}{P(X)}$$

Where:

- **$P(C|X)$**  = Probability that the waste belongs to class C after observing sensor data X.
- **$P(X|C)$**  = Probability of observing the sensor values if the waste belongs to class C.
- **$P(C)$**  = Prior probability of class C.
- **$P(X)$**  = Probability of the observed sensor data.

The algorithm calculates this probability for each class (Recyclable and Non-Recyclable). The class with the highest probability becomes the final prediction.

## Sensor Data Used for Waste Classification

A smart waste segregation system collects information from several sensors before making a prediction.

### 1. Weight Sensor

Measures the weight of the waste object.

Examples:

- Plastic bottle – Light weight
- Glass bottle – Heavy
- Metal can – Medium to heavy

Weight helps distinguish between different waste materials.

### 2. Moisture Sensor

This sensor detects whether the waste is wet or dry.

Examples:

- Food waste → High moisture
- Plastic → Low moisture
- Paper → Moderate moisture

Wet waste is generally non-recyclable, while dry waste has higher recycling potential.

### **3. Metal Sensor**

A metal detector identifies metallic objects.

Examples:

- Aluminum cans
- Steel containers
- Iron objects

If metal is detected, the object is usually recyclable.

### **4. Color Sensor or Optical Sensor**

This sensor recognizes the color and surface characteristics of the waste.

Examples:

- Transparent bottle
- Brown cardboard
- Green glass bottle
- Colored plastic

Optical information improves classification accuracy.

### **5. Additional Sensors (Optional)**

Modern systems may also use:

- Infrared Sensor
- Camera
- Ultrasonic Sensor
- Temperature Sensor

These sensors provide additional information for better classification.

## Working Process of Naïve Bayes in Waste Classification

The complete process consists of several stages.

### Step 1: Data Collection

Sensors collect data from every waste object.

Example:

| Sensor   | Reading |
|----------|---------|
| Weight   | 0.45 kg |
| Moisture | Low     |
| Metal    | Yes     |
| Color    | Silver  |

This information forms the input dataset.

### Step 2: Data Preprocessing

The collected data is cleaned before training.

This includes:

- Removing incorrect values
- Handling missing data
- Normalizing sensor readings
- Converting categorical values into machine-readable form

A clean dataset improves prediction accuracy.

## Step 3: Model Training

The Naïve Bayes classifier is trained using historical waste data.

Example training data:

| Weight | Moisture | Metal | Color       | Class          |
|--------|----------|-------|-------------|----------------|
| Light  | Low      | No    | Transparent | Recyclable     |
| Heavy  | High     | No    | Brown       | Non-Recyclable |
| Medium | Low      | Yes   | Silver      | Recyclable     |
| Heavy  | High     | No    | Mixed       | Non-Recyclable |

The algorithm learns the probability of each feature belonging to each class.

## Step 4: Probability Calculation

When a new waste object arrives, the algorithm calculates:

- Probability of being recyclable
- Probability of being non-recyclable

Example:

Sensor readings:

- Weight = Medium
- Moisture = Low
- Metal = Yes
- Color = Silver

Calculated probabilities:

- Recyclable = 95%
- Non-Recyclable = 5%

Therefore, the object is classified as **Recyclable**.

## Step 5: Waste Segregation

After prediction, the smart bin automatically directs the waste into the appropriate container.

For example:

- Plastic Bottle → Recyclable Bin
- Food Waste → Non-Recyclable Bin
- Metal Can → Recyclable Bin

Automatic segregation improves recycling efficiency.

## Example Classification

Suppose a smart dustbin receives an object with the following sensor readings:

| Sensor   | Reading         |
|----------|-----------------|
| Weight   | Light           |
| Moisture | Low             |
| Metal    | No              |
| Color    | Transparen<br>t |

The Naïve Bayes classifier compares these values with its training data.

The calculated probabilities may be:

- Recyclable = 92%
- Non-Recyclable = 8%

Final Prediction:

**Recyclable**

Now consider another object:

| Sensor | Reading |
|--------|---------|
| Weight | Heavy   |

Moisture    High

Metal        No

Color        Mixed

Predicted probabilities:

- Recyclable = 18%
- Non-Recyclable = 82%

Final Prediction:

**Non-Recyclable**

## Advantages of Using Naïve Bayes

The Naïve Bayes classifier offers several benefits for smart waste management systems:

- Simple and easy to implement.
- Fast training and prediction.
- Requires less computational power.
- Works well with small and medium-sized datasets.
- Handles multiple sensor inputs effectively.
- Provides accurate classification for many real-world problems.
- Suitable for real-time waste segregation systems.
- Reduces manual effort and operational costs.

## Limitations

Despite its advantages, Naïve Bayes has some limitations.

- Assumes all sensor features are independent.
- Performance depends on the quality of training data.
- May produce lower accuracy when sensor readings are highly correlated.
- Complex waste materials may require advanced machine learning models.

Even with these limitations, Naïve Bayes remains an excellent choice for simple and efficient waste classification systems.

# Applications

Naïve Bayes-based waste classification systems can be used in:

- Smart cities
- Government buildings
- Schools and colleges
- Shopping malls
- Airports
- Railway stations
- Hospitals
- Industrial waste management plants
- Public recycling centers

These applications improve waste segregation and increase recycling efficiency.

## Contribution to Sustainable Development Goals (SDGs)

This system directly supports several SDGs:

### **SDG 11 – Sustainable Cities and Communities**

- Promotes cleaner public spaces.
- Reduces landfill waste.
- Improves urban waste management.

### **SDG 12 – Responsible Consumption and Production**

- Encourages recycling.
- Minimizes resource wastage.
- Supports sustainable production practices.

### **SDG 13 – Climate Action**

- Reduces greenhouse gas emissions from landfills.
- Promotes environmentally friendly waste disposal.

## Conclusion

Naïve Bayes is an efficient and reliable machine learning algorithm for classifying recyclable and non-recyclable waste using sensor data. By collecting information from sensors such as weight, moisture, metal, and color sensors, the system calculates the probability of each waste item belonging to a specific category. The waste is then automatically sorted into the appropriate bin, improving recycling efficiency and reducing environmental pollution. Its simplicity, speed, and low computational requirements make it suitable for real-time smart waste management systems. Implementing Naïve Bayes in smart bins and recycling facilities contributes to cleaner cities, better resource utilization, and the achievement of **SDG 11 (Sustainable Cities and Communities)** and **SDG 12 (Responsible Consumption and Production)**.